

Midterm - Creating Images with Diffusion Models

Analysis Report / Assessment Questions

ITAI 2376 | Team DataProxy | Fashion-MNIST

Team	DataProxy
Course	ITAI 2376
Dataset	Fashion-MNIST
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Project Summary

Team DataProxy used Fashion-MNIST to study how a diffusion model learns to generate clothing images. The notebook adds noise to 28x28 grayscale images, trains a U-Net to predict that noise, and then uses the reverse process to remove noise step by step. The report below contains the team's assessment responses and observations from the project.

1. Understanding Diffusion

Jonah Joseph

Explain what happens during the forward diffusion process, using your own words and referencing the visualization examples from your notebook.

The forward diffusion process slowly adds noise to the image over several steps. At the start, the image can be seen clearly, but over time as noise is added, it gets more blurry. This makes details from the original image become unnoticeable and almost look like a different image altogether.

Why do we add noise gradually instead of all at once? How does this affect the learning process?

We add noise gradually to train the model on how the image looks at different noise levels. This allows the model to learn how to remove noise at each step. If the noise was added all at once, it would be harder for the model to adjust and learn how to remove it.

Look at the step-by-step visualization - at what point (approximately what percentage through the denoising process) can you first recognize the image? Does this vary by image?

The image starts to become recognizable around 60% of the way through the denoising process. It can vary by image because some images have more noticeable details than others. It also depends on how well the model is able to remove noise from the image.

2. Model Architecture

Samantha Mireles

Why is the U-Net architecture particularly well-suited for diffusion models? What advantages does it provide over simpler architectures?

U-Net works well because diffusion models need to turn noisy images into clearer images step by step. Its encoder reduces the image's resolution to learn larger patterns, such as overall shapes. Its decoder increases the resolution again to produce a detailed output, often a prediction of the noise to remove. This allows U-Net to use both the big picture and small details. Compared with a simpler network, it can combine information at different scales while keeping track of where details belong.

What are skip connections and why are they important? Explain them in relation to our model.

Skip connections are shortcuts that carry information from earlier layers directly to later layers. In a U-Net, they connect encoder layers to decoder layers at matching resolutions. This gives the decoder access to details that could otherwise be lost during downsampling, such as edges and object positions. In our diffusion U-Net, the decoder combines these details with broader patterns learned deeper in the network to make more accurate noise predictions.

Describe in detail how our model is conditioned to generate specific images. How does the class conditioning mechanism work?

Class conditioning tells the model which category of image to generate. The class label is converted into a numerical embedding, and the timestep is also encoded so the model knows which stage of denoising it is handling. The U-Net uses both pieces of information while predicting the noise. During generation, the chosen class label is supplied repeatedly while the model starts from random noise and denoises it step by step.

3. Training Analysis

Elijah Raines

What does the loss value of your model tell us?

The loss tells us how far the model's predicted noise is from the actual noise that was added to the image. A lower loss means the model is getting better at recognizing and removing noise. For example, our training loss started around 0.1873 and later dropped to about 0.115, which showed that the model was improving as it trained.

How did the quality of your generated images change throughout the training process?

As training continued, the images started becoming less random and more structured because the model was getting better at predicting the noise. Early in training, the generated images were more unclear, but as the loss decreased, the model had a better chance of creating shapes that looked more like the Fashion-MNIST clothing classes. We still needed to fully check the final generated images, but the decreasing loss showed that the model was learning.

Why do we need the time embedding in diffusion models? How does it help the model understand where it is in the denoising process?

The time embedding tells the U-Net what step of the diffusion process it is currently working on. An image at an early timestep may only have a little noise, while an image at a later timestep may be almost completely noisy. The time embedding helps the model know how much denoising it should perform. For example, removing noise at Step 90 is different from removing noise at Step 10, so the model needs that timestep information to make the correct prediction.

4. CLIP Evaluation

Fifehanmi Ogedengbe

What do the CLIP scores tell you about your generated images? Which images got the highest and lowest quality scores?

Response to be completed.

Develop a hypothesis explaining why certain images might be easier or harder for the model to generate convincingly.

Response to be completed.

How could CLIP scores be used to improve the diffusion model's generation process? Propose a specific technique.

Response to be completed.

5. Practical Applications

Brandon Matias

How could this type of model be useful in the real world?

Synthetic Data Generation: We can use this approach to generate synthetic images, such as clothing, handwritten digits, or medical scans, to train downstream models when real datasets are scarce, expensive, or restricted by privacy laws.

E-Commerce & Apparel Design: By using class conditioning, design teams and retailers could quickly prototype apparel styles, test ideas, and preview product concepts before production.

What are the limitations of our current model?

Low Resolution and Grayscale Only: Our network is limited to 28x28 single-channel images, while many real applications need much higher-resolution RGB outputs.

Rigid Conditioning and Linear Noise Schedule: The model uses a simple linear noise schedule and fixed class labels, which limits the range and flexibility of what it can generate.

If you were to continue developing this project, what three specific improvements would you make and why?

1. Switch to a Cosine Noise Schedule & DDIM Sampler: This could preserve more image detail and reduce the number of generation steps.

2. Integrate CLIP Text-Prompt Conditioning: This could allow generation from flexible natural-language prompts instead of only fixed numeric class IDs.

3. Third improvement: *The provided source does not include a third completed improvement response.*

Submission Note

This report is organized from the team's provided assessment-question document. The CLIP Evaluation responses and the third improvement in Practical Applications were not completed in the source, so they are clearly marked instead of being filled in with unsupported information.